GAME DESIGN & LEVEL SPECIFICATION DOCUMENT (GDD)

An Echo Has No Shape

Single-Verb First-Person Acoustic Exploration & Epistemic Puzzle Labyrinth

Author / Role	Kevin (Lefi) Shan — Solo Game & Level Designer, Programmer	**Engine & Stack**	Unity 6 (URP), C#, Custom HLSL Spherical Wave Shaders
Production Scope	4-Week Solo Prototype · USC CTIN (Situated Role: The Traveler)	**Core Constraints**	1 Unchanging Verb (Press E) · Zero HUD / Minimaps · Single ~550m Scene

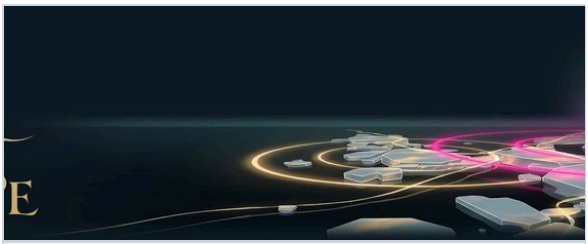


Figure 1.1: Player's white echo pulse colliding with the Siren's magenta wave in the Glacial Ocean.

Executive Summary & Design Goal

Most echolocation games treat sound as a binary flashlight in the dark: once the pulse reveals the room's geometry, the mechanic stops evolving because the relationship between player input and returned information remains static.

To focus a 4-week solo cycle entirely on level design and systemic puzzle variation, An Echo Has No Shape locks the player to a single unchanging verb (Press E to Cast Echo, plus F to Attune) across a continuous ~550m × 550m sightless basin. Progression is driven not by unlocking new abilities, but by subverting the epistemic rules of how each region answers back.

1. Core Design Pillars & Theoretical Framework

Section 1.1 – 1.3

1.1 World as Presence (The Traveler)

In Peter Brinson's taxonomy, the Traveler differs from the Investigator in how meaning is constructed: rather than piecing together forensic clues about a past event, the Traveler experiences a world in the present tense through movement, attention, and pause. Where the player walks and when they stop to listen determine what the world reveals.

1.2 Mechanical Restraint (1 Verb)

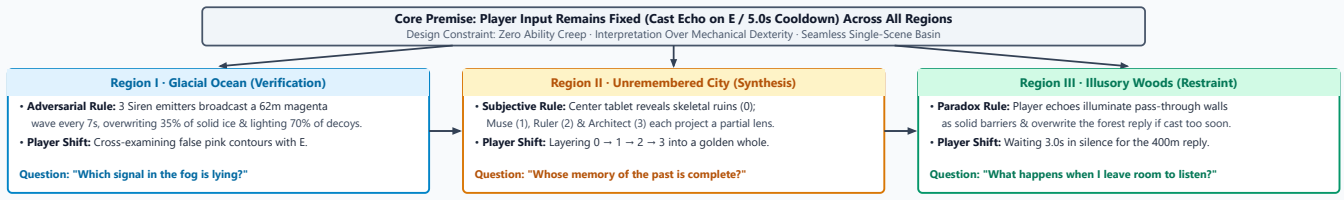
The player's action set never expands: Cast Echo (E, 42m range, 5.0s cooldown) and Attune (F at stone tablets). Eliminating ability creep, combat, and inventory management forces every regional puzzle to recontextualize how the player interprets the returning acoustic signal.

1.3 Three Epistemic Shifts

The title is literal: an echo has no intrinsic shape of its own; it borrows the contour of whatever surface it strikes. Across the three regions, the player's relationship to their own voice shifts from skeptical verification (Glacial Ocean), to bearing witness (Unremembered City), to relinquishing control (Illusory Woods).

Figure 1.2 — System Architecture: Three Epistemic Inversions Under a Single Unchanging Input (Press E)

Calibration (Hub) → Verification (Region I) → Synthesis (Region II) → Restraint (Region III)



2. Regional Epistemic Specification Matrix ("Rule of Three")

Table 2.1 — Regional Rulesets

Specification Axis	Hub: Stone Monument	Region I: Glacial Ocean	Region II: Unremembered City	Region III: Illusory Woods
Epistemic Stage	**Calibration** Baseline sonar orientation	**Verification** Cross-examining adversarial noise	**Synthesis** Layering biased historical lenses	**Relinquishing Control** Pausing in silence to hear reply
Habit Subverted	Relying on permanent sight and HUD waypoints.	Treating every lit contour in the mist as solid geometry.	Assuming a single observer holds the whole truth.	Spamming E automatically on cooldown (every 5.0s).
System Counterpart	**Prophecy Tablet** Static central anchor	**Siren Emitters (×3)** 62m magenta pulse every 7.0s	**Muse, Ruler & Architect** 150m–200m perspective waves	**Forest Reply Center** 3.0s delayed 400m green reply
Failure / Friction	Initial disorientation in open volumetric mist.	Stepping on a pink decoy floe submerges & respawns player.	Out-of-order attunement triggers a 45m/s collapse wave.	Pressing E early triggers a 0.35s overwrite & shuts walls.
Diegetic Tablet Hint	*"Three silences wait in the mist..."*	*"Not every voice across the ice wishes you to reach the shore."*	*"The city stands whole within the meeting of many eyes."*	*"The woods have opened to the voice you left room to hear."*

3. Seamless World Topology & Four-Scale Diegetic Wayfinding

`~550m × 550m Acoustic Basin (Main.unity)`



Figure 3.1 — Master World Map (~550m × 550m): Single-scene layout showing the central Stone Monument Hub, nested inner/outer regional fog bounds, outskirts tablets, and acoustic guide splines.

3.1 Hub-and-Spoke Single-Scene Layout

To enforce zero HUD markers, minimaps, or loading screens, the entire ~550m × 550m basin resides inside one continuous Unity scene around the Stone Monument Hub. Players awaken at the monument, attune to the prophecy tablet to ignite initial guide splines, and traverse the three regions in any non-linear order:

- **NW Basin — Glacial Ocean:** Descends from the western shore across branching ice floes toward the northern terminal tablet and sky beacon.
- **NE Plateau — Unremembered City:** Crosses the West Bridge causeway into a 200m × 220m street grid enclosed by an outer reset boundary.
- **Southern Valley — Illusory Woods:** Accessible from both the NW hub spline (Tablet T1) and SE city approach (Tablet T2), terminating at the SW Sentinel Tower.

3.2 Dual-Collider Atmospheric & Audio Crossfades

Without a visible horizon line, regional transitions rely on diegetic atmospheric and musical crossfades. Each region is wrapped in nested OuterBounds and InnerBounds trigger volumes. As the player traverses the outskirts buffer (normalized distance $t \in [0, 1]$), the environment controller smoothly interpolates volumetric fog color, fog attenuation distances (D_start, D_end), and regional music stems.

Figure 3.2 — Non-Linear Hub Progression & 3/3 Golden Gate Finale Aggregator

Any-Order Regional Completion → 450m World Reverberation + Sky Beacons I/II/III



3.3 Four-Scale Sightless & Diegetic Wayfinding Specification

`Table 3.1 — Spatial Orientation Stack`

Spatial Scale	Effective Range	Diegetic Mechanism	System & Shader Implementation	Navigational Function
1. Macro Scale	`150m – 450m`	**World Reverberations & Sky Beacons**	Regional completion triggers a 450m golden wave (12s hold) and ignites BeaconGlacier/City/Woods.	Provides basin-wide landmarks across the mist, drawing the player back to the Hub.
2. Meso Scale	`50m – 150m`	**Dual-Collider Fog & Music Crossfades**	Interpolates volumetric fog color, falloff, and music stems between OuterBounds and InnerBounds.	Communicates regional threshold crossings before specific geometry becomes visible.
3. Micro Scale	`15m – 50m`	**Acoustic Guide Splines**	SplineContainer ribbons remain dormant until struck by an echo, then send particles along path tangent.	Preserves active discovery: paths answer back only when the player calls out.
4. Immediate Scale	`0m – 15m`	**Seen vs. Unseen Material Split**	Splits assets into Seen (faint silhouettes in mist) and Unseen (0% base visibility until echoed).	Ensures primary anchors (tablets, statues, spires) remain readable between cooldowns.



Figure 3.3 — Hub Initial State (0/3): White player echo sweeping across the monolith ring and prophecy tablet.



Figure 3.4 — Hub Completion State (3/3): Unlocked Golden Gate, active guide splines, and all three sky beacons.

4.1 Region I: Glacial Ocean — Adversarial Acoustic Signals & Verification

`Epistemic Role: Skeptical Verification`

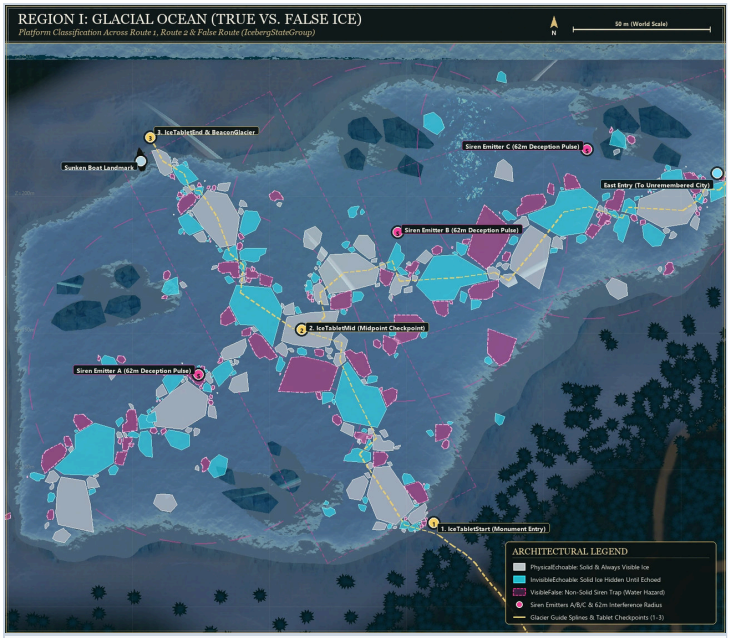


Figure 4.1 — Region I Level Design Map (Glacial Ocean): Showing permanent anchor islands (white), hidden solid stepping stones (cyan), non-solid Siren decoy traps (magenta dashed), and 62m Siren emitter radii.

Design Problem: Platforming Tension Without Dexterity Frustration

First-person jumping across ice floes becomes routine if the player's echo is 100% reliable, and frustrating if jumps demand pixel-tight dexterity. The Glacial Ocean places the tension in evaluating which signal to trust. Three unseen Siren Emitters (A/B/C) broadcast a 62m magenta pulse every 7.0s ± 1.0s that corrupts nearby ice visibility and lures hasty players onto non-solid water traps.

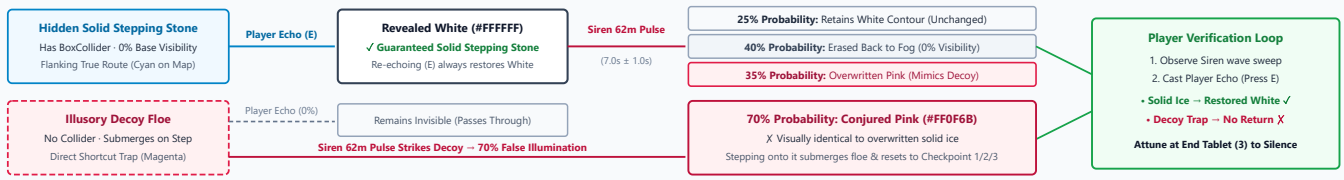
Three-Tier Iceberg Collision & Visibility Specification

- **1. Solid Anchor Islands (White on Map):** Permanent islands and tablet checkpoints (1 Start, 2 Twin Spires Midpoint, 3 End) that remain faintly visible in the mist as safe observation platforms.
- **2. Hidden Solid Stepping Stones (Cyan on Map):** Physically solid floes with 0% base visibility. Only the player's white echo reliably reveals them in pure white (#FFFFFF); when struck by a Siren wave, 40% are erased back to fog and 35% are overwritten in pink.
- **3. Illusory Decoy Floes (Magenta Dashed on Map):** Non-solid traps with no collider (submerging on contact). The player's white echo passes straight through them (0% return), but the Siren's pulse falsely illuminates 70% of decoys in bright pink (#FF0F6B).

Spatial Layout & Shortcut Temptation Loop

Between Checkpoints 1 → 2 → 3, non-solid decoy floes are arranged along tempting direct shortcuts, while true solid stepping stones curve along flanking arcs. Because a Pink iceberg could be either a 35% overwritten solid floe or a 70% non-solid decoy, players must wait for the Siren wave to pass and cast their own E echo to verify before jumping.

Figure 4.2 — Glacial Ocean State Machine: Probabilistic Iceberg Visibility & Player Cross-Examination Player White Echo (100% Ground Truth) vs. Siren 62m Magenta Pulse (7.0s ± 1.0s Cycle)



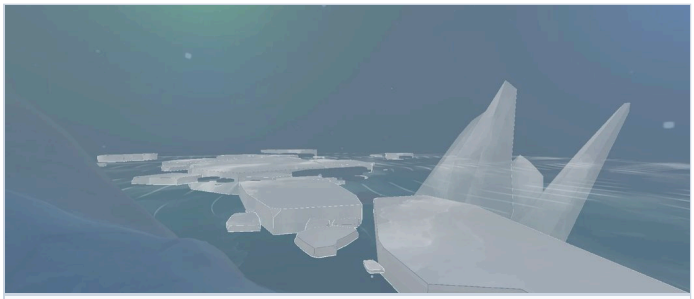


Figure 4.3 — White Echo Verification: Player-cast white echo revealing guaranteed solid stepping stones before the twin ice spires.



Figure 4.4 — Golden Route Resolution: Attuning at the terminal tablet silences the Sirens and illuminates the route in gold.

4.2 Region II: Unremembered City — Multi-Perspective Memory Synthesis

`Epistemic Role: Bearing Witness & Synthesis`

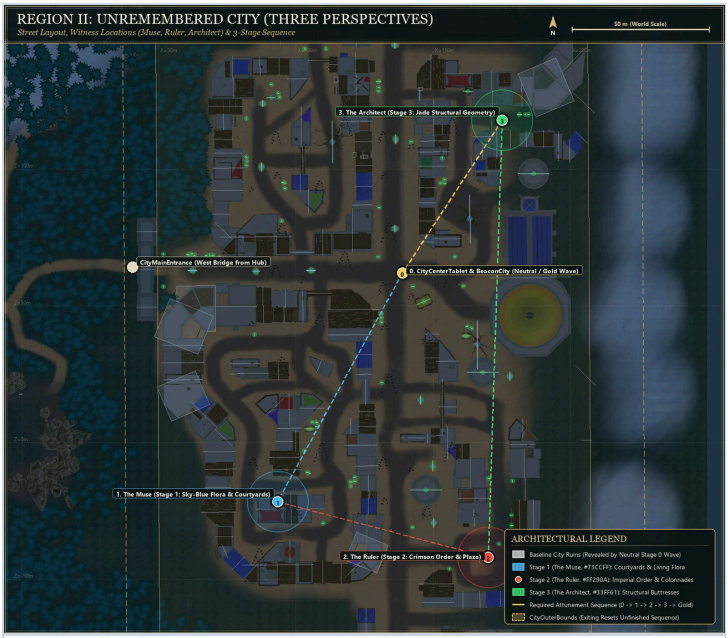


Figure 4.5 — Region II Level Design Map (Unremembered City): Showing the central plaza tablet (0) and the three witness quarters—The Muse (1, SW), The Ruler (2, SE), and The Architect (3, NE)—plus the outer reset boundary.

Design Problem: Environmental Storytelling Without Dialogue Trees

How do you express the theme that history is a collection of biased, incomplete projections over a ruin purely through level geometry and state? No single observer's memory restores the city; only by traversing all three quarters in sequence (0 → 1 → 2 → 3) and layering their partial truths together does the city become whole.

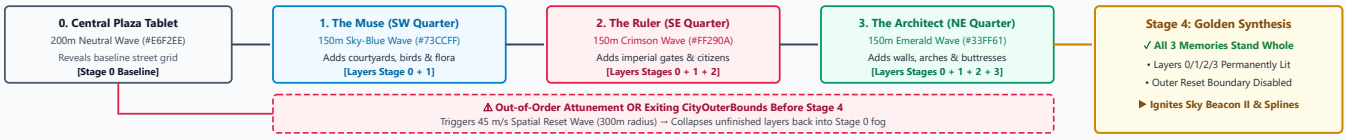
Five-Stage Cumulative Architecture Stack (Stage 0 → 4)

- **Stage 0 · Central Plaza Tablet (Neutral Wave #E6F2EE):** Emits a 200m neutral pulse revealing the skeletal street grid while leaving courtyards, gates, and structural spans absent.
- **Stage 1 · SW Quarter — The Muse (Sky-Blue Wave #73CCFF):** Remembers the city as a sanctuary of art and nature; shifts fog to cerulean and materializes courtyard structures, birds, and flora.
- **Stage 2 · SE Quarter — The Ruler (Crimson Wave #FF290A):** Remembers the city as an instrument of authority; shifts fog to crimson and layers monumental gates, colonnades, and citizen silhouettes.
- **Stage 3 → 4 · NE Quarter — The Architect (Emerald #33FF61 → Gold #FFCC4D):** Restores load-bearing walls, arches, and buttresses, locking all layers into permanent Stage 4 Golden Synthesis.

Sequence Enforcement & 45m/s Spatial Reset Wave

Attuning to a witness statue out of sequence (anything other than 0 → 1 → 2 → 3) or stepping outside CityOuterBounds before completing the synthesis triggers a 45m/s Spatial Reset Wave from the central plaza that collapses unfinished projections back into fog.

Figure 4.6 — Unremembered City State Machine: 0 → 1 → 2 → 3 Cumulative Layering & 45m/s Collapse Loop Each Valid Attunement Preserves Prior Architectural Layers Toward Stage 4 Synthesis



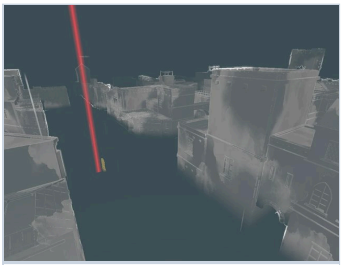


Figure 4.7 — Stage 0 (Neutral): Baseline skeletal ruins and central plaza beacon.



Figure 4.8 — Stage 1 (The Muse): Sky-blue fog, sanctuary courtyards, and flora.



Figure 4.9 — Stage 2 (The Ruler): Crimson fog, imperial archway, and citizens.



Figure 4.10 — Stage 4 (Synthesis): All three perspective layers restored in gold.

4.3 Region III: Illusory Woods — Pass-Through Labyrinth & 3.0s Silence Tuning

`Epistemic Role: Relinquishing Control`



Figure 4.11 — Region III Level Design Map (Illusory Woods): Pass-through illusory barriers (bright green bars) block arterial junctions until dissolved by the 400m forest reply wave after 3.0s of silence.

The Timing Paradox: Your Own Echoes Sustain the Barrier

Inside the dirt-corridor labyrinth, 8 pass-through illusory barriers (FalseWalls) sit across the primary junctions leading toward the southwest Sentinel Tower. In the forest, the player's own echoes maintain the barrier: casting a white echo (42m range, 1.4s duration) illuminates these illusory tree walls alongside real corridor walls, making direct routes appear blocked.

Systemic Resolution: When the player casts an echo and then waits in silence for 3.0s, the center of the forest (WoodsEchoSource) answers back with a 400m Verdant Reply Wave (#47FF6B at 50m/s) that dissolves all illusory barriers for 5.0s. Pressing E during the hold window triggers a 0.35s Active Wave Overwrite that carves through the green wave and snaps the barriers shut.



Figure 4.12 — Guide Spline: Pine corridor lit by an acoustic guide spline.

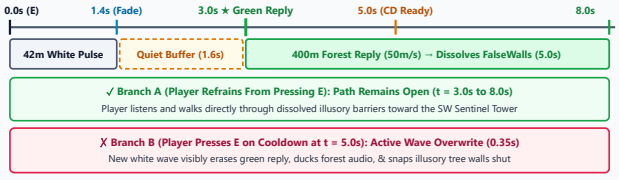


Figure 4.13 — Sentinel Tower: Clearing revealed in emerald reply glow.

Table 4.1 — Playtest Telemetry: 3.0s Silence Delay (6 Cohorts)

Delay	Accidental	Intentional	Observed Playtest Behavior
`1.0s`	85%	15%	Players trigger green reply while walking without noticing rule.
`2.0s`	45%	50%	Mixed recognition; rapid walkers trigger by accident.
`3.0s` ★	**10%**	**90%**	**[Tuned Sweet Spot] Pulse fades at 1.4s; leaves 1.6s quiet buffer before 5.0s cooldown.**
`4.0s`	5%	60%	Impatient players begin pressing E near end of cooldown.
`5.0s`	0%	20%	Matches cooldown: players press E on cooldown, resetting timer.
`6.5s`	0%	5%	Too long: players assume nothing happened and keep echoing.

Figure 4.14 — The 8-Second Call-and-Response Cadence & 0.35s Overwrite Systemic Cause & Effect



5. Design Retrospective & Key Takeaways

`Post-Mortem Synthesis`

5.1 Epistemic Variety Over Ability Creep

Keeping the player on a single E input meant progression had to come from recontextualizing how returned information is read—shifting from verification (Glacial Ocean), to synthesis (Unremembered City), to restraint (Illusory Woods).

5.2 Systemic Feedback Beats Text Hints

In the Illusory Woods, tablet poetry alone did not break the habit of echoing on cooldown. Introducing the 0.35s active wave overwrite (where a new player pulse visibly erases the forest's green wave and snaps walls shut) raised intentional discovery from 15% to 90%.

5.3 Aligning Level Topology With Rules

Each region's layout reinforces its acoustic rule: decoy floes along tempting direct shortcuts in the Glacial Ocean, three witnesses separated across distinct city quarters, and pass-through barriers across arterial forest junctions.